DOS
LAB 2

Outputs :

1) Front :

```
Welcome to Bazar.com CLI!
Type a command: search <topic>, info <item_number>, purchase <item_number>, or exit
> search distributed systems
[CACHE] MISS - fetching http://catalog1:5001/search/distributed systems
Response: [{'item_number': 1, 'title': 'How to get a good grade in DOS in 40 minutes a day'}, {'item_number': 2, 'title': 'RPCs for Noobs'}]
> info 3
[CACHE] MISS - fetching http://catalog2:5001/info/3
Response: {'cost': 25, 'item_number': 3, 'stock': 12, 'title': 'Xen and the Art of Surviving Undergraduate School'}
> info 3
[CACHE] HIT for /info/3
Response: {'cost': 25, 'item_number': 3, 'stock': 12, 'title': 'Xen and the Art of Surviving Undergraduate School'}
> purchase 1
[FRONTEND] Sending purchase request to http://order1:5002/purchase with payload {'item_number': 1}
{'message': 'Order placed', 'status': 'success'}
> info 1
[CACHE] MISS - fetching http://catalog2:5001/info/1
Response: {'cost': 55, 'item_number': 1, 'stock': 119, 'title': 'How to get a good grade in DOS in 40 minutes a day'}
> info 3
[CACHE] HIT for /info/3
Response: {'cost': 25, 'item_number': 3, 'stock': 12, 'title': 'Xen and the Art of Surviving Undergraduate School'}
> info 1
[CACHE] HIT for /info/1
Response: {'cost': 55, 'item_number': 1, 'stock': 119, 'title': 'How to get a good grade in DOS in 40 minutes a day'}
> info 3
[CACHE] EXPIRED for /info/3
[CACHE] MISS - fetching http://catalog1:5001/info/3
Response: {'cost': 25, 'item_number': 3, 'stock': 12, 'title': 'Xen and the Art of Surviving Undergraduate School'}
>
```

2)Admin :

```
=== ADMIN MENU ===

1. Update Stock
2. Update Cost
3. Update Both
0. Exit
Enter choice: 1
Item number: 1
New stock: 100
[ADMIN] Response from http://order1:5002: 200 - {'message': 'Book updated', 'status': 'success'}

1. Update Stock
2. Update Cost
3. Update Both
0. Exit
Enter choice: 2
Item number: 1
New cost: 100
[ADMIN] Response from http://order1:5002: 200 - {'message': 'Book updated', 'status': 'success'}

1. Update Stock
2. Update Cost
3. Update Both
0. Exit
Enter choice: 3
Item number: 5
New stock: 200
New cost: 200
[ADMIN] Response from http://order1:5002: 200 - {'message': 'Book updated', 'status': 'success'}
```

3)Order and Catalog (Backend Servers):

```
catalog2-1  | 172.18.0.5 - - [16/Jun/2025 18:33:13] "POST /update/1 HTTP/1.1" 200 -
order2-1    | 172.18.0.7 - - [16/Jun/2025 18:33:13] "POST /admin_update/1 HTTP/1.1" 200 -
catalog1-1  | 172.18.0.4 - - [16/Jun/2025 18:33:20] "POST /update/1 HTTP/1.1" 200 -
order1-1    | 172.18.0.7 - - [16/Jun/2025 18:33:20] "POST /admin_update/1 HTTP/1.1" 200 -
catalog1-1  | 172.18.0.4 - - [16/Jun/2025 18:33:27] "POST /update/2 HTTP/1.1" 200 -
order1-1    | 172.18.0.7 - - [16/Jun/2025 18:33:27] "POST /admin_update/2 HTTP/1.1" 200 -
catalog1-1  | 172.18.0.4 - - [16/Jun/2025 18:33:32] "POST /update/3 HTTP/1.1" 200 -
order1-1    | 172.18.0.7 - - [16/Jun/2025 18:33:32] "POST /admin_update/3 HTTP/1.1" 200 -
catalog1-1  | 172.18.0.6 - - [16/Jun/2025 18:33:44] "GET /search/distributed%20systems HTTP/1.1" 200 -
catalog2-1  | 172.18.0.6 - - [16/Jun/2025 18:33:49] "GET /info/1 HTTP/1.1" 200 -
catalog1-1  | 172.18.0.4 - - [16/Jun/2025 18:33:53] "POST /decrease_stock/1 HTTP/1.1" 200 -
order1-1    | 172.18.0.6 - - [16/Jun/2025 18:33:53] "POST /purchase HTTP/1.1" 200 -
catalog1-1  | 172.18.0.6 - - [16/Jun/2025 18:33:56] "GET /info/2 HTTP/1.1" 200 -
catalog2-1  | 172.18.0.6 - - [16/Jun/2025 18:34:01] "GET /info/3 HTTP/1.1" 200 -
catalog2-1  | 172.18.0.5 - - [16/Jun/2025 18:34:03] "POST /decrease_stock/7 HTTP/1.1" 200 -
order2-1    | 172.18.0.6 - - [16/Jun/2025 18:34:03] "POST /purchase HTTP/1.1" 200 -
```

```
catalog1-1  | 172.18.0.4 - - [16/Jun/2025 18:28:45] "POST /update/5 HTTP/1.1" 200 -
order1-1    | 172.18.0.9 - - [16/Jun/2025 18:28:45] "POST /admin_update/5 HTTP/1.1" 200 -
order1-1    | 172.18.0.9 - - [16/Jun/2025 18:28:45] "POST /admin_update/5 HTTP/1.1" 200 -
order1-1    | 172.18.0.9 - - [16/Jun/2025 18:29:08] "POST /admin_update/1 HTTP/1.1" 200 -
order1-1    | 172.18.0.9 - - [16/Jun/2025 18:29:24] "POST /admin_update/5 HTTP/1.1" 200 -
```